GAHAN GANDI

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SUMMARY

- Versatile software engineer with a Master's in Computer Science from the University of Missouri. Proficient in programming languages including C, C++, Python, Java, JavaScript, and TypeScript, with extensive experience in web technologies such as React, Angular, HTML, CSS, Node.js, and Express. Skilled in developing full-stack applications, data analytics, machine learning, and computer vision solutions, and experienced in using tools and platforms like AWS, Docker, and Git. Proven track record of enhancing system performance, optimizing data processing, and deploying scalable applications.

EDUCATION

University Of Missouri, Columbia, MO, USA — Aug 2022 – May 2024
Master of Computer Science, GPA: 3.9/4.0
- ***Relevant Coursework:*** *Web Application Development II, Databases and Analytics, Supervised Learning, Logic in Computer Science, Data Mining and Information Retrieval, Design and Analysis of Algorithms II, Digital Image Processing, Computer Vision*

Indian Institute Of Technology Mandi, Mandi, HP, India — Aug 2018 – Jun 2022
Bachelor of Technology in Computer Science, GPA: 3.6/4.0
- ***Relevant Coursework:*** *Data Structures and Algorithms, Computer Organization, Data Science, Machine Learning, Pattern Recognition, Neural Networks, Map Reduce, Computing Systems for Data Processing*

SKILLS & OTHER

Programming Languages: C, C++, Python, Java, JavaScript, TypeScript
Web Technologies: React, Angular, HTML, CSS, Node.js, Express
Databases: MySQL, MongoDB, NoSQL, Firebase, Relational Databases
Tools & Platforms: AWS, Git, Docker, Rest API, OpenCV, PyTorch, Nvidia DeepStream, Linux, Unix
Areas of Expertise: Front End, Back End, Full Stack, Computer Vision, Software Development, web Application development, Machine Learning, data structures and algorithms, object oriented design

PROFESSIONAL EXPERIENCE

Department of Transportation, University of Missouri, Columbia, MO, USA — Aug 2022 – Present
Junior Software Engineer
- Built a React app with JavaScript and TypeScript to visualize dynamic traffic and road data across Missouri, enhancing decision-making processes.
- Designed interactive dashboards using React with REST APIs for real-time traffic, weather, and transportation data, improving data accessibility by 25%.
- Crafted app's backend in Python with a NoSQL database, prioritizing robust data management and user authentication, reducing security risks by 40%.
- Established a CI/CD pipeline using GitHub Actions to automate code integration and deployment to Firebase, resulting in 50% faster deployment times.
- Managed and deployed the app using AWS and Docker, ensuring consistent performance across all development stages, improving uptime by 99.5%.

SAYeTech, Columbia, MO, USA — July 2024 – Present
Software Development Engineer
- Building a web application for a startup using React, Javascript and typescript.
- Designed beautiful admin dashboards and graphs for visualization using React JS Charts and React MUI Charts
- Integrated the application with firebase for authentication and backend, and integrated Google Maps and OpenWeatherAPI into the application.
- Designing a Mobile application Using React native.

Y STEM and Chess Inc, Columbia, MO, USA — Jun 2024 – July 2024
Software Developer
- Migrated app's front-end from Angular to React for enhanced UI responsiveness and back-end from a legacy system to Node.js, improving scalability.
- Handled complex back-end development tasks by integrating the Node.js back-end with Stockfish chess engine to facilitate real-time gameplay.

University of Missouri, Columbia, MO, USA — Jan 2023 – Apr 2024
Graduate Research Assistant, Computer Vision
- Created a PyTorch-based computer vision app for vehicle and lane detection on Linux, boosting detection accuracy by 20% and enhancing road safety.
- Fine-tuned and retrained the YOLOPv2 model on Nvidia DeepStream, improving video analytics frame rate by 30% through optimized C++ code.
- Built a real-time video processing system in Node.js to flag potential collisions from truck camera feeds, reducing collision incidents by 15% for MoDOT.
- Deployed complex video analytics solutions on AWS using Docker, streamlining CI/CD processes for efficient and scalable real-time analytics.
- Developed Java and MongoDB back-end solutions, enhancing data transaction speeds by 50% for real-time vehicle detection and analytics.
- Developed Camera Calibration techniques to compute distances of vehicles following the TMA Truck using the camera, without using LIDAR thus reducing cost of the product.
- Implemented a real time video processing solution for the Missouri Department of Transportation, analyzing live streams from cameras on the back of TMA Truck to flag vehicles at high risk of collision, enhancing construction crew safety on work zones.

University of Missouri, Columbia, MO, USA — Nov 2022 – Jan 2023
Student Research Assistant
- Designed an interactive UI using Plotly Dash and React for uploading and visualizing engineering data that significantly boosted research efficiency.
- Optimized backend operations using Python, significantly enhancing data processing and analysis capabilities for a civil engineering research lab.
- Implemented data storage solutions with Python and MySQL, ensuring robust management of complex A-Scan and B-Scan data from CSV/Excel files.

Indian School of Business, Chandigarh, India — Dec 2019 – Jan 2020
Software Intern
- Worked a geo spatial analysis using Python, JavaScript and ML algorithms to process satellite images in Google Earth Engine, classifying areas into urban, forest, and agricultural zones, enabling real-time analysis of decade-long satellite data.

PROJECTS

Enrollment System Enhancement (Link)
- Utilized Angular, Node, Express, and MongoDB to create a robust student enrollment app, resulting in an enhanced UX and operational efficiency.

Optimized Research Tool

- Built a tool in React to analyze Griewank Benchmark Function runtimes across devices, aiding postdoctoral research with improved data accessibility.

Enhanced Game Development (Link)

- Engineered a Card Memory Game using Angular and TypeScript, which improved interactive engagement and user retention.

Advanced Expense Management (Link)

- Created an Expense Tracker app with React and Firebase, integrating user authentication and streamlining financial tracking via cloud deployment.

Innovative Disease Detection (Link)

- Applied image processing techniques and CNN with ResNet-50 for plant disease classification, supporting agricultural advancements.

Crime Data Analysis (Link)

- Constructed a comprehensive database for the Chicago crime dataset, enhancing data retrieval and analysis capabilities for public safety research.

Theoretical Math Application (Link)

- Devised Fermulerpy, a Python package for Number Theory, incorporating key functions to advance mathematical research and education.

CERTIFICATIONS

- **Amazon Web Services (AWS) Certified Cloud Practitioner (Link)**
- **Become a Front End Developer from Educative**